

Speaker notes — rehearsal cues

51 frames · » advance a fragment · ▲ say this one verbatim · **bold** is a number or name to get exactly right

Act 0 — the setup · 1–7

1 title

- thanks — committee, for reading and being here
- **3 years**; trustworthy non-Gaussian inference for weak-lensing cosmology

2 the picture

- hot, almost uniform, small density fluctuations
- expands; fluctuations grow under gravity → galaxies, clusters, cosmic web

3 Λ CDM

- **six** free parameters
- assumptions: GR · homogeneous+isotropic on large scales · inflation, near-Gaussian, adiabatic · baryons + CDM + Λ
- ▲ extremely successful — six numbers fit everything to remarkable precision
- » but not satisfactory: phenomenological not first-principles · **95 %** is two dark components · tensions between probes

4 the probes

- several probes test a model like that
- » ▲ ours is gravitational lensing

5 strong and weak

- light bends passing the mass on the way to us → distorted image
- dramatic = arcs, rings, multiple images — **strong**, very rare
- ▲ **weak** happens everywhere: every galaxy slightly changed → how matter, incl. dark, is distributed

6 Euclid

- launched **2023**, taking data now
- **billions of galaxies, a third of the sky**, ~order of **magnitude** more statistical power
- » ▲ signal encodes growth of structure + geometry since the Big Bang
- but needs sophisticated algorithms and statistics — that is this thesis

7 the chain

- analyses are incredibly complex; this is the last steps, the scientific part
- shapes in → mass map → summary statistics → compare to theory/sims → posterior
- » ▲ first half, first two papers = **making the map**
- » ▲ second half, last two papers = everything after it
- every step can bias or distort → trustworthy needs quantified, calibrated, unbiased

Act 1 — does the map matter? · 8–20

8 Part 1 divider

- first paper — but to motivate it, back to weak-lensing basics

9 shear and convergence

- convergence = isotropic magnification; shear = anisotropic stretching
- convergence **not directly observable**
- ▲ lensing is **coherent**, intrinsic shapes **random** → average many galaxies, random cancels, shear remains

10 what convergence is

- **scalar** field (shear is spin-2) — easier; most HOS only defined on κ
- **projected matter** along the line of sight, weighted by lensing efficiency
- → convergence maps = **mass maps**
- real example: **UNIONS**

11 same potential

- both are second derivatives of the **lensing potential** (projection of the gravitational potential)
- Fourier space → simple linear relation

12 exact relation, imperfect data

- irregular sampling · noise **much larger than the shear** · masks
- → **ill-posed inverse problem**: several visibly different solutions fit the data
- » ▲ need an assumption about what κ looks like

13 Kaiser–Squires

- linear inversion applied directly; **thirty years**, almost every survey, for simplicity
- » amplifies shape noise at small scales · mask leakage across the whole map (non-local operator)
- pure KS almost unreadable → Gaussian filtering in practice, at the cost of small scales

14 data term + regulariser

- every method = optimisation: **data fidelity** + **regulariser (the prior)**
- data term is the **same for all**; what changes is the prior on κ and how it is solved

15 MCALens

- state of the art among **model-driven** (hand-crafted prior)
- κ = **Gaussian** (Wiener filter) + **non-Gaussian** (the peaks)
- **alternating minimisation** — solve one holding the other, iterate to convergence
- » each sub-problem is a **proximal step**: gradient towards the shear, then operator pulling back to the prior

16 the question

- until this paper methods were compared on **map quality** — MSE against the truth
- but the map is not the final product; we care about the **parameters**
- » practical for Euclid: is an advanced method **worth the effort**, or is any reasonable one good enough?

17 the experiment

- everything **held fixed** except the mass-mapping method
- **cosmo-SLICS**, Euclid-like, whole chain shear → posterior per method
- details come back in Parts 3 and 4
- » the only thing that varies is the reconstruction

18 the three maps

- baselines: **KS**, and **KS + inpainting** (fills the mask to match the field statistically, meant to cut leakage)
- these two are what **Euclid** plans — mass mapping treated as preprocessing, so the simplest became default
- **MCALens** is the state of the art
- so what posteriors do they give?

19 the answer *first corner plot, take your time*

- explain the plot: off-diagonal = joint distribution of a pair; diagonal = one parameter; two shades = **68 and 95 %**; smaller = more precise
- **inpainting makes no difference** — grey and red almost identical, grey hard to see behind the red
- **MCALens visibly tighter on every parameter**
- stems: both are improvement over KS. **open** = RMSE, **4 %** better. **filled** = constraining power, **×2.6, 160 %**
- map quality and constraining power are **not the same objective**
- → surveys should use advanced reconstruction to maximise scientific output

20 where the gain comes from

- add smaller and smaller scales: KS **saturates below 8'**; MCALens keeps improving
- MCALens recovers the **small-scale structure** KS loses to noise

21 Part 2 divider

- reconstruction matters → motivation to build the best one we can
- this paper led by **Hubert Leterme**

22 what we want

- obvious place to look is **deep learning** — tried, and it works
- **four things**: accurate · flexible (same model when noise or mask changes) · fast · fast uncertainty quantification
- **none of the existing methods have all four**

23 plug-and-play

- from the optimisation literature; inverse problem with a **learned** prior, very flexible
- start from the Part 1 iteration: gradient step on the data term, then proximal step
- PnP replaces the proximal operator with a **deep denoiser** trained on simulations → converges to a **fixed point**
- better than analytical priors: the denoiser learns a far more complex, realistic prior
- **Swin transformer**, 7 million parameters, trained once on white Gaussian noise over a range of levels
- ▲ noise covariance and mask enter **only in the data term, at inference** → same network for any noise, any footprint. That is the flexibility
- walk-through: two shear components, κ initialised at zero → **forward** (gradient) gives a noisy map → **backward** (denoiser) projects onto realistic mass maps
- » output fed back, iterate to convergence
- *short path if behind: eight iterations, alternating data step and denoiser; residual variant slightly better, in the backup*

Act 3 — the summaries · 25–44

25 the second half

- what comes after the map: the **summary statistic**, and inference from it
- so far the statistic was **held fixed on purpose** — nothing that moved could be blamed on it. Now it is the subject

26 compression

- map is ~**a hundred thousand correlated pixels** — very hard to use directly
- » compress to a low-dimensional summary
- » the choice determines how much information is kept

24 accurate, calibrated

- accurate maps **and** error bars, **perfectly calibrated, with mathematical guarantees**
- via **moment networks + conformalized quantile regression**
- axes: x = reconstruction error, y = size of calibrated bars → **lower left is better**
- colour = miscoverage: circles above target before, every diamond on it after. Not *who covers* — everyone covers, that is the guarantee — but who covers with the **tightest bars**
- ▲ accuracy within ~1 % of **DeepMass**, trained for that exact mask and noise; **smallest calibrated bars** of all
- » DeepMass needs retraining per mask/noise → not applicable to a real survey; PnPMass trained **once**, any field

27 two-point

- standard: **two-point correlation function** — $\xi_{\pm}$ in real space, power spectrum in harmonic
- » every flagship result of the decade — **KiDS, DES, HSC** — is two-point: analytic prediction, understood covariance, two decades of systematics work
- » ▲ for a **Gaussian** field the power spectrum is **sufficient**. The late-time field is not Gaussian

28 same power spectrum

- collapse and non-linear growth → haloes and filaments around voids
- these two completely different fields have **the same power spectrum** — two-point cannot tell them apart
- → need **higher-order**: peaks, wavelets, ℓ_1 -norm, Minkowski functionals

29 the gain

- forecasts on simulations: adding HOS to the power spectrum **significantly improves** the constraints

30 wavelets *the parenthesis*

- Fourier uses sines and cosines; wavelets are **localised, oscillating, dilated and translated**
- coefficient at scale a , position b = how much structure of size a sits at b
- » suited to convergence maps — localised structures with a characteristic size and position. Small wavelet → small-scale structure; large → large-scale
- each band covers a different frequency range → study **scale by scale**, largely independent

31 the two statistics

- both start from the wavelet decomposition, isotropic wavelet called the **starlet**
- transform → set of **band-pass images**, each a characteristic angular size
- » count peaks on each band, histogram their amplitudes = **wavelet peak counts**
- » sensitive to non-Gaussian features; peaks ↔ dark matter haloes, whose abundance depends strongly on cosmology
- » but counting peaks throws away voids, filaments, the rest → sum the **absolute values** of all coefficients per band, histogram those = **starlet ℓ_1 -norm**
- » think of it as a **PDF of the field in the wavelet domain**

32 to the parameters

- compare the statistic with predictions under different cosmologies
- to carry all the uncertainties: **Bayesian statistics**

33 classical inference

- Bayes: posterior $\propto$ **likelihood** $\times$ **prior**
- classical writes the likelihood **analytically**, usually Gaussian in the data vector → needs a mean prediction and a covariance. Works for the power spectrum
- » sample with **MCMC**
- for peaks or ℓ_1 : **neither** an analytic mean **nor** a Gaussian likelihood → different route

34 generative modelling

- unknown distribution, but we have samples from it → fit a parametric model
- once trained: generate new samples, **and evaluate the density**
- » the family behind image generation; look how far they have come

35 normalizing flows

- start from a unit Gaussian → target, through **simple invertible transformations**, each a neural network
- invertible → density from the **change-of-variables formula**; trained to maximise the likelihood of the samples
- the flow learns from samples how to turn a Gaussian into the target

36 simulation-based inference

- no analytic likelihood, but **we have a simulator** — parameters in, simulated data out, with all the stochasticity
- draw from the prior, run, keep the **(parameter, data) pairs**
- » those train a **conditional** flow → the flow is the posterior. Give it the real observation, get the posterior

37 Part 3 divider

- **third question**: how much information do the statistics keep, and can HOS extract *all* of it?

38 why build a statistic by hand

- ℓ_1 beats the power spectrum — » but that is "easy". Real question is **how close to all of it**
- we already use networks for inference — why not learn the compression too?
- **reasons to be careful**: needs a lot of training data · hard to interpret · generalises less predictably — a feature in the sims but not the data biases the result unnoticed
- the theoretically optimal way: build the network into the SBI pipeline, train it with the flow — **VMIM**

39 what optimal means

- network = **compressor**, maps → low-dimensional summary; flow maps summary → posterior
- » trained together to maximise the **mutual information** between summary and parameters $\equiv$ maximising the expected log posterior
- » so in theory it estimates the **most information any summary of these maps can carry**
- → compare our analytical HOS against that

40 the setup

- most of the work in this paper is in this figure — the whole SBI pipeline, maps to parameters, plus both the analytical statistics and the neural compressor (many architectures, always VMIM)
- the two approaches share **everything apart from the summary**

41 the gap

- first result: not far apart, but the CNN wins by **36 %** in FoM → ℓ_1 loses part of the information
- » but the comparison is **not symmetric** — a property of these maps the ℓ_1 does not yet use

42 tomography *eight clicks, one per build step*

- sources sliced into **redshift bins**
- » most distant bin » lensed by all the matter in front » gives its map » a nearer bin » shorter column » another map » one map per bin, **not independent** — the same structure appears in several
- how the signal changes bin to bin says where the matter sits along the line of sight → how structure grew
- a **per-bin** statistic cannot see it

43 two routes

- two-point extends easily — cross-bin correlations. For HOS there is no obvious equivalent
- **first**: extra maps for the cross structure — multiply each pair of bins pixel by pixel; strong only where **both** have structure at the same place. Compute the ℓ_1 on those too
- » **second**: change the statistic — generalise the ℓ_1 from one map to a **pair** of bins
- each pixel has a coefficient in each map → instead of one histogram per map (the two curves on the axes) » one **two-dimensional** histogram over the pair
- **diagonal** cells = both maps equally strong; **off-diagonal** = one strong, one weak

44 the answer

- same maps, same flow, four summaries: per-bin ℓ_1 » + product cross-maps » joint ℓ_1 » CNN — falls almost perfectly on the joint ℓ_1
- network was trained to be information-optimal → the **joint ℓ_1 essentially encodes all the cosmologically useful information**
- and with **no training** — efficient, interpretable, robust

Act 4 — baryons · 45–50

45 Part 4 divider

- HOS work well in idealised cases → final paper, a **messy, realistic** scenario

46 the collision of scales *longest beat, 2:13*

- systematics = in the measurement, not in the model; instrumental or physical. Unmodelled → biased result
- most groups check the **data vector**. Not enough — what matters is the effect on the **posterior**
- dominant astrophysical one at small scales: **baryonic feedback** — AGN and supernovae push gas out of haloes, suppressing small scales
- our inference sims are **dark-matter-only**, and feedback models disagree → cannot model reliably
- the problem: non-Gaussian information and contamination live on **the same small scales** → conservative answer is to cut them
- » *optimistic*: only the smallest scales contaminated, most of the information survives
- » *pessimistic*: contamination reaches further, and after the cut the power spectrum does just as well
- two questions: how much does unmodelled feedback bias HOS, and how does it grow with area — and after the cut is there anything left over the power spectrum

47 the pipeline

- same SBI pipeline as the previous paper
- » maps at different cosmologies » Euclid-like noise and masks » starlet transform » data vector conditions a flow
- one difference

48 how large the bias is

- **CosmoGrid** gives every realisation with and without feedback → train the flow on **dark-matter-only** (no baryon model at all), feed it a **baryonified** observation
- run across survey areas to see how the bias evolves
- previous-generation area: not significant. Errors shrink with area, systematic does not → **Euclid: really bad**; full sky worse
- ▲ HOS more biased than the power spectrum — more sensitive to the contaminated small scales

49 the cuts

- remove small scales until the bias is under **0.3σ**
- power spectrum: lower $\ell_{\max}$ — **860 at 2,000 deg²** down to **340 at full sky**
- wavelets: drop bands — **the finest band alone** is enough at every area

50 is there anything left

- Stage IV posteriors, **baryon-safe scales only**: » power spectrum » peak counts » ℓ^1 -norm
- ▲ ℓ^1 reaches $\times 1.8$ the power spectrum's FoM; $\times 2.6$ at full sky where the wavelet cut fits better
- peaks comparable to the power spectrum but **complementary** — different degeneracy directions in the w_0 planes
- ▲ HOS are not only deep non-linear probes — the signal survives on **quasi-linear** scales with **no baryon model at all**
- and we can do better: the cut is not optimised, and as feedback modelling improves the analysis moves back into the non-linear regime, where the gain is larger
- *if pressed: $\times 1.8$ carries ± 0.6 — "about a factor of two". Not significant at $2,000 \text{ deg}^2$, significant from $5,000 \text{ up}$. Peaks $\times 1.07$ Stage IV, $\times 1.17$ full sky*

[Close](#) · 51

51 conclusions

- followed the chain from shapes to parameters: making the map, extracting information, simulation-based inference
- **map**: the choice of method is **not neutral** — it changes the FoM significantly. And **PnPMass** — physics + deep learning, accurate, calibrated bars, trained once for any mask and noise
- **summary**: the **joint ℓ^1 -norm** extracts as much as a neural compressor trained to be optimal, **without any training**
- **systematics**: HOS still give significant additional information after cutting the baryon-contaminated scales
- ▲ steps usually treated as neutral choices **do** change the result. With calibrated and tested methods we extract more, and **trust what we get out**
- *if nulling came up: and the same joint reading is what makes redshift nulling survivable for a higher-order analysis, which is in the thesis*
- thank you